

ESSLLI 'Semantic properties and combinatorial restrictions of attitude predicates'

Day 2

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A recap and a preview

Yesterday

- Correlation between semantic and combinatorial properties
- An account in terms of meaning-driven selectional restriction.
 - Embedding predicates are type-wise compatible with all sorts of complements.
 - Some combinations result in triviality $\leadsto$ ungrammaticality.

Today

- A close look at a particular account of this form, focusing on the **hope-wh* puzzle (non-veridical preferentials $\sim$ anti-rogrativity).
- A preliminary experimental evidence, with discussion of methodological challenges and potential counterexamples.

Today's roadmap

1. A potential meaning-driven combinatorial property: **hope-wh*
2. Uegaki & Sudo's (2019) account
3. An experimental assessment
4. Summary and teasers for the following days

**A potential meaning-driven
combinatorial property: **hope-wh***

Veridicality and preferentiality

Veridicality

A predicate V is **veridical** iff $\llbracket \exists x \text{ Vs that } p \rrbracket$ entails that p .

Representational vs. preferential

- An attitude predicate is **representational** if it quantifies over the attitude holder's information state (e.g. beliefs).
- An attitude predicate is **preferential** if it involves comparison of alternatives based on the attitude holder's preference.

(Bolinger 1968; Heim 1992; Villalta 2008; Anand & Hacquard 2013)

	Representational	Preferential
Veridical	<i>know, forget</i>	<i>like, hate, surprise, annoy</i>
Non-veridical	<i>be certain, believe</i>	<i>hope, wish, fear</i>

Non-veridical + preferential → anti-rogative

Generalization (Uegaki & Sudo 2019)

All **non-veridical preferential** predicates are anti-rogative.

- (1) John **knows** $\left\{ \begin{array}{l} \checkmark \text{that Max will sing} \\ \checkmark \text{who will sing} \end{array} \right\}$. [+ver,rep]
- (2) John **is certain** $\left\{ \begin{array}{l} \checkmark \text{that Max will sing} \\ \checkmark \text{about who will sing} \end{array} \right\}$. [-ver,rep]
- (3) John **likes/hates** $\left\{ \begin{array}{l} \checkmark \text{that Max will sing} \\ \checkmark \text{who will sing} \end{array} \right\}$. [+ver,pref]
- (4) John **hopes/fears** $\left\{ \begin{array}{l} \checkmark \text{that Max will sing} \\ *(*\text{about}) \text{ who will sing} \end{array} \right\}$. [-ver,pref]

...but not vice versa.

Uegaki & Sudo's account

Unified semantics of complementation

Unified semantics of complementation

- All clause-embedding predicates select for a *question*.
⇒ Interrogative-embedding.
- Declarative complements denote a *singleton question*.
⇒ Declarative-embedding.

(5) $\llbracket \text{John knows who danced} \rrbracket = 1$
iff $\llbracket \text{know} \rrbracket(\text{john}, \llbracket \text{who danced} \rrbracket)$

(6) $\llbracket \text{John knows that Ann danced} \rrbracket = 1$
iff $\llbracket \text{know} \rrbracket(\text{john}, \{\llbracket \text{Ann danced} \rrbracket\})$

(Ciardelli et al. 2015; Uegaki 2015; Theiler et al. 2016)

Denotations of representational predicates

$$(7) \quad \llbracket \text{be certain} \rrbracket^w = \lambda Q_{\langle st, t \rangle} \lambda x. \exists p \in Q[\mathbf{B}_w(x, p)]$$

$$(8) \quad \llbracket \text{know} \rrbracket^w = \lambda Q_{\langle st, t \rangle} \lambda x : \underline{\exists p \in Q[p(w)]}. \exists p \in Q[\mathbf{K}_w(x, p)]$$

- All clause-embedding predicates are *typewise* compatible with both types of complements.
- Selectional restrictions are explained by lexical semantic contents.

(Ciardelli & Roelofsen 2015; Uegaki 2015; Theiler et al. 2017; Mayr 2017)

Focus sensitivity of preferentials

Scenario: Lisa knows that syntax will be taught this semester. She does not want syntax to be taught by John, since she believes that he is not a good syntax teacher. Also, she wants syntax to be taught on Tuesdays (regardless of who teaches it), since that would fit her schedule. Lisa learns that John will teach syntax on Tuesdays.

(9) Lisa is happy that John will teach syntax on **TUES**days.
(Judgment: *true*)

(10) Lisa is happy that **JOHN** will teach syntax on Tuesdays.
(Judgment: *not true*)

(After Heim 1992; Villalta 2008; Romero 2015)

Degree-based analysis of preferentials

(11) $\llbracket \text{be happy}_C \rrbracket^w(p)(x)$ (after Romero 2015)

- is defined only if $p(w) = 1$ and x believes p in w .
- if defined, is true iff x 's preference for p (in w) is stronger than the 'standard' of x 's preferences for alternatives in C .

(12) $\llbracket \text{be happy}_C \rrbracket^w$
 $= \lambda p \lambda x : [p(w) \wedge \mathbf{B}_w(x, p)].$
 $\text{Pref}_w(x, p) > \theta(\{\text{Pref}_w(x, p') \mid p' \in C\})$

- $\text{Pref}_w(x, p) :=$ the degree of x 's preference for p in w .
- $\theta(D_{dt}) :=$ the standard given the degrees in D

Preferentials in the unified semantics of complementation

$$(13) \quad \llbracket \text{be happy}_C \rrbracket^w(Q)(x)$$

- is defined only if there is $p \in Q$ s.t. $p(w) = 1$ and x believes p in w .
- if defined, is true iff there is $p' \in Q$ s.t. $p'(w) = 1$, x believes in p' (in w) and x 's preference for p' (in w) is stronger than the 'standard' of x 's preferences for alternatives in C .

$$(14) \quad \llbracket \text{be happy}_C \rrbracket^w$$

$$= \lambda Q \lambda x : \exists p \in Q [p(w) \wedge \mathbf{B}_w(x, p)].$$

$$\exists p' \in Q \left[\begin{array}{c} p'(w) \wedge \mathbf{B}_w(x, p') \wedge \\ \text{Pref}_w(x, p') > \theta(\{\text{Pref}_w(x, p'') \mid p'' \in C\}) \end{array} \right]$$

Illustration: veridical preferential *be happy*

$$(15) \quad \llbracket \text{John is happy}_C \text{ about } [\mathbf{who\ sang}] \sim C \rrbracket = \\ \lambda w : \exists p \in Q[p(w) \wedge \mathbf{B}_w(\mathbf{j}, p)].$$

$$\exists p' \in Q \left[\begin{array}{c} p'(w) \wedge \mathbf{B}_w(\mathbf{j}, p') \\ \text{Pref}_w(\mathbf{j}, p') > \theta(\{\text{Pref}_w(\mathbf{j}, p'') \mid p'' \in Q\}) \end{array} \right]$$

‘There is an answer $p' \in Q$ s.t. p' is true, John believes p' , and the degree to which John prefers p' is greater than the standard given Q .’

$$(16) \quad \llbracket \text{John is happy}_C \text{ that } [[\mathbf{Ann}]_F \text{ sang}] \sim C \rrbracket \\ = \lambda w : \exists p \in \{A\}[p(w) \wedge \mathbf{B}_w(\mathbf{j}, p)].$$

$$\exists p' \in \{A\} \left[\begin{array}{c} p'(w) \wedge \mathbf{B}_w(\mathbf{j}, p') \\ \text{Pref}_w(\mathbf{j}, p') > \theta(\{\text{Pref}_w(\mathbf{j}, p'') \mid p'' \in Q\}) \end{array} \right]$$

‘ A is true and John believes A ; the degree to which John prefers A is greater than the standard given Q .’

Contrasting veridical and non-veridical preferentials

(17) John **is happy** that Ann sang.

⇒ Ann sang.

⇒ John believes that Ann sang.

veridicality

belief

(18) John **hopes** that Ann sang.

≠ Ann sang.

≠ John believes that Ann sang.

~~veridicality~~

~~belief~~

Non-veridical preferential predicates

Semantics of veridical preferentials

$$(19) \quad \llbracket \text{be happy}_C \rrbracket^w = \lambda Q \lambda x: \exists p \in Q[p(w) \wedge \mathbf{B}_w(x, p)].$$

$$\exists p' \in Q \left[\begin{array}{c} p'(w) \wedge \mathbf{B}_w(x, p') \wedge \\ \text{Pref}_w(x, p') > \theta(\{\text{Pref}_w(x, p'') \mid p'' \in C\}) \end{array} \right]$$

Semantics of non-veridical preferentials

$$(20) \quad \llbracket \text{hope}_C \rrbracket^w = \lambda Q \lambda x: \not\exists p \in Q[p(w) \wedge \mathbf{B}_w(x, p)].$$

$$\exists p' \in Q \left[\begin{array}{c} \not p'(w) \wedge \not \mathbf{B}_w(x, p') \wedge \\ \text{Pref}_w(x, p') > \theta(\{\text{Pref}_w(x, p'') \mid p'' \in C\}) \end{array} \right]$$

Additional assumptions about θ

$\theta(D_{dt}) :=$ the standard degree given D

Threshold significance

Gradable predicates in general presuppose that there is a degree $d \in D$ such that $d > \theta(D_{dt})$

$\approx$ A predication with a gradable predicate makes sense only if there is an entity in the comparison class whose degree along the relevant scale exceeds the threshold.

- (21) **Situation:** there is no student such that John wants them to sing.
- a. #John doesn't hope that ALICE will sing.
 - b. #John isn't happy about which student will sing.

Deriving triviality: hope+wh

$$\begin{aligned}(22) \quad & \llbracket \text{John hopes}_C (\text{about}) [\text{who left}] \sim C \rrbracket \\ & = \lambda w. \exists p \in \{A, B, C\} \\ & \quad [Pref_w(\mathbf{j}, p) > \theta(\{Pref_w(\mathbf{j}, p') \mid p' \in \{A, B, C\}\})]\end{aligned}$$

Presupposition of θ John prefers either A , B or C to a degree higher than the standard.

Assertion There is an answer in $\{A, B, C\}$ such that John prefers it to a degree higher than the standard.

$\Rightarrow$ Whenever the presupposition is met, the assertion is true.

$\Rightarrow$ Gajewski's (2002) L-triviality (or its presup. extension).

$\Rightarrow$ Ungrammaticality

Deriving non-triviality: be happy+wh

$$\begin{aligned} (23) \quad & \llbracket \text{John is happy}_C \text{ about [who left]} \sim C \rrbracket \\ & = \lambda w. \exists p \in \{A, B, C\} [p(w) \wedge \mathbf{B}_w(\mathbf{j}, p) \wedge \text{Pref}_w(\mathbf{j}, p) > \\ & \quad \theta(\{\text{Pref}_w(\mathbf{j}, p') \mid p' \in \{A, B, C\}\})] \end{aligned}$$

Presupposition of θ John prefers either A , B or C to a degree higher than the standard.

Assertion There is an answer in $\{A, B, C\}$ such that (i) it is true, (ii) John believes it, and (iii) John prefers it to a degree higher than the standard.

$\Rightarrow$ (Even) if the presupposition is met, the assertion can be true or false due to the veridical (and doxastic) restriction on existential quantification.

$\Rightarrow$ Non-triviality

Deriving non-triviality: hope+that

$$\begin{aligned} (24) \quad & \llbracket \text{John hopes}_C \text{ that } \llbracket \text{Ann} \rrbracket_F \text{ left} \rrbracket \sim C \\ & = \lambda w. \exists p \in \{A\} [\text{Pref}_w(\mathbf{j}, p) > \theta(\{ \text{Pref}_w(\mathbf{j}, p') \mid p' \in \{A, B, C\} \})] \end{aligned}$$

Presupposition of θ John prefers either A , B and C to a degree higher than the standard.

Assertion There is an answer in $\{A\}$ such that John prefers it to a degree higher than the standard.

$\Rightarrow$ (Even) if the presupposition is met, the assertion can be true or false given the singleton domain (i.e., $\{A\}$) of existential quantification, originating from the declarative complement.

$\Rightarrow$ Non-triviality

Recapping the pattern

- (25) $\llbracket \text{John is happy}_C \text{ that } \llbracket [\text{Ann}]_F \text{ left} \rrbracket \sim C \rrbracket =$
 $\lambda w. \exists p \in \{A\} [p(w) \wedge \mathbf{B}_w(\mathbf{j}, p) \wedge \text{Pref}_w(\mathbf{j}, p) >$
 $\theta(\{\text{Pref}_w(\mathbf{j}, p') \mid p' \in \{A, B, C\}\})] \quad \rightsquigarrow \text{Non-trivial}$
- (26) $\llbracket \text{John is happy}_C \text{ about } \llbracket \text{who left} \rrbracket \sim C \rrbracket$
 $= \lambda w. \exists p \in \{A, B, C\} [p(w) \wedge \mathbf{B}_w(\mathbf{j}, p) \wedge \text{Pref}_w(\mathbf{j}, p) >$
 $\theta(\{\text{Pref}_w(\mathbf{j}, p') \mid p' \in \{A, B, C\}\})] \quad \rightsquigarrow \text{Non-trivial}$
- (27) $\llbracket \text{John hopes}_C \text{ that } \llbracket [\text{Ann}]_F \text{ left} \rrbracket \sim C \rrbracket$
 $= \lambda w. \exists p \in \{A\} [\text{Pref}_w(\mathbf{j}, p) > \theta(\{\text{Pref}_w(\mathbf{j}, p') \mid p' \in$
 $\{A, B, C\}\})]$
 $\rightsquigarrow \text{Non-trivial}$
- (28) $\llbracket \text{John hopes}_C \text{ (about) } \llbracket \text{who left} \rrbracket \sim C \rrbracket$
 $= \lambda w. \exists p \in \{A, B, C\}$
 $[\text{Pref}_w(\mathbf{j}, p) > \theta(\{\text{Pref}_w(\mathbf{j}, p') \mid p' \in \{A, B, C\}\})] \quad \rightsquigarrow \text{Trivial}$

An account of the generalization that non-veridical preferentials are anti-rogative.

- A general template for preferential predicates: there is a proposition p in Q that is preferred.
- **Veridical predicates**: the true proposition is preferred. $\leadsto$ non-trivial
- **Non-veridical predicates**: some proposition is preferred $\leadsto$ trivial, given threshold significance.

An experimental assessment

Experimentally testing the semantics-selection correlations

The theory predicts correlations between semantic properties and selectional properties across predicates. This can be tested by behavioral experiments.

Categorical vs. gradient judgment

- Above, we assumed that the relevant semantic properties (e.g. veridicality, preferentiality) are **categorical** in nature.
- But, it is plausible that native speakers' *perception* about these notions are **gradient** (White & Rawlins 2018).
- How should we test our theory in view of the more realistic, gradient perception of the relevant semantic properties by speakers?

Chemla, Homer, & Rothschild (2011) on NPIs

- **Chemla et al. 2011:** perceived, gradient strength of DE-ness of various grammatical environments correlates with the acceptability of NPIs in those environments.
- **Structural parallel between the two problems:**
 - Acceptability of NPI — DE-ness of the environment
 - Acceptability of a clause type — lexical semantics of the embedding predicate

Hypothesis

Speakers' judgment about selectional restrictions is rooted in the **perceived semantic property** of embedding predicates.

U&S's prediction

Perceived degrees of non-veridicality and preferentiality of predicates affect the acceptability of interrogative clauses in their complement position.

Exp 1: Introduction

U&S's prediction

Perceived degrees of non-veridicality and preferentiality of predicates affect the acceptability of interrogative clauses in their complement position.

- In order to test this prediction, in Exp1, I preselected 87 preferential predicates in English, and tested their
 - (a) veridicality; and
 - (b) acceptability with interrogative complementsusing a web-based experiment.
- If the prediction is correct, there will be a correlation between (a) and (b).

Exp 1: The method of predicate selection

- Following the proposal by Villalta 2008, I used focus-sensitivity as a proxy for preferentiality.
 - **Method of predicate selection:**
 1. Take Villalta's (2008) list of Spanish focus-sensitive predicates, and add her English translations to the set. In doing this, those predicates that don't seem to be focus-sensitive according to the test below are excluded.
 2. Add synonyms of the predicates in 1 according to *Oxford American Writer's Thesaurus*, again keeping in mind that only those predicates that satisfy the below test are added.
- (29) Alice **V**ed that she would teach SYNTAX on Tuesdays, but she didn't **V** that she would teach syntax on TUESdays.
- a. consistent $\leadsto$ **V** is focus-sensitive.
 - b. inconsistent $\leadsto$ **V** isn't focus-sensitive.

Exp 1: 87 Predicates

(30) *hope, desire, prefer, fear, admire, be bored, be glad, accept, be fascinated, like, care, be interested, regret, be worried, be sorry, complain, be surprised, advise, encourage, recommend, suggest, want, wish, demand, expect, hate, love, anticipate, long, yearn, favor, dread, be terrified, be panicked, applaud, praise, approve, appreciate, respect, be exhausted, be entertained, be captivated, be absorbed, be enchanted, be pleased, fancy, be happy, enjoy, be concerned, be troubled, be bothered, mind, be amused, be intrigued, be attracted, be struck, welcome, be disturbed, be distressed, be upset, be stressed, be strained, be unsettled, protest, grumble, whine, lament, criticize, be astonished, be amazed, be startled, be astounded, be stunned, be flabbergasted, be staggered, be shocked, miss, crave, urge, persuade, press, pressure, advocate, loath, dislike, detest, despise*

Exp 1: 2-block design

■ Veridicality block

Suppose your friend Susie tells you the following:

"I demanded that my mother was at home."

Hearing this, how confident would you be in concluding that Susie's mother was at home?

1 (Not confident at all)	2	3	4 (Neutral)	5	6	7 (Very confident)
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■ Interrogative block:

How natural does the following sentence sound to you?:

"I demanded which students were present at the party."

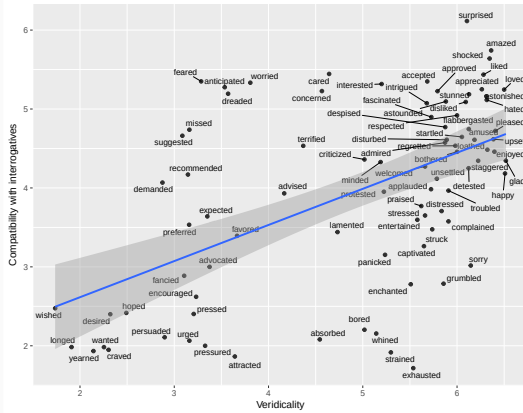
1 (Completely unnatural)	2	3	4 (Neutral)	5	6	7 (Completely natural)
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- The frame **w/o a preposition** was used regardless of the pred's syntactic category (incl. e.g., *be glad*, *be surprised*).

Exp 1: Items and participants

- In both blocks, there was one trial for each of the 87 predicates. Each participant answered 194 questions in total for the whole experiment. No fillers.
- Both the order of the two blocks and the order of the predicates in each block were randomized.
- 70 English native speakers were recruited through Prolific.ac.
- The participants were paid £1.5.

Results



- Significant effect of Veridicality on Compatibility with interrogatives ($p = 0.00154$; mixed-effects logistic regression).
- Whether the predicate is a passive/adjectival form showed neither main effect nor interaction with Veridicality.

Exp 1: Discussion

- The significant main effect of **veridicality** confirms U&S's prediction that veridicality affects compatibility with interrogative complements within the set of focus-sensitive predicates.
- **Concern:** the set of focus-sensitive predicates are preselected prior to the experiment. To evaluate the hypothesis, ideally, we would like to **experimentally measure focus-sensitivity/preferentiality** in a larger sample of predicates, and examine if focus-sensitivity/preferentiality + veridicality affects the compatibility with interrogative complements.

Exp 2: Introduction

- Exp 2 evaluated the hypothesis using datasets from the **MegaAttitude project** (White & Rawlins, 2016; 2018).
- The datasets contain cloud-sourced **selectional patterns of 1,000 English verbs** as well as **veridicality judgment of 517 English verbs** (between subjects).
- The datasets do not encode data about preferentiality or focus-sensitivity *par se*. However, it contains information that may correlate with it: **S-lifting** (??):

- (31) a. It is raining, I believe.
b. *It is raining, I was surprised.

Exp 2: Method

- The datasets provide the selection/veridicality data for the active and passive voice of each verb. For each verb-voice pair, I extracted data on its compatibility with *which*-complements (**which**), its compatibility with S-lifting (**slift**), and veridicality (**veridicality**).
- **slift** has been argued to (at least roughly) correlate with preferentiality/focus-sensitivity. Assuming that this correlation is indeed there, our current hypothesis predicts that **veridicality** together with **slift** affects **which**.

Exp 2: Analysis

- After aggregating the data across participants, I ran in R a simple linear logistic regression on **which**, with **voice** (baseline: active), **veridicality**, **slift** and interactions between them as independent variables.

(32) **which** ~ **veridicality** * **voice** * **slift**

Exp 2: Results

	β	Std. Error	t-value	p
(Intercept)	1.58285	6.99138	0.226	0.821
VERIDICALITY	0.19875	2.73996	0.073	0.942
[VOICE: passive]	0.25098	9.32178	0.027	0.979
SLIFT	1.12908	1.86009	0.607	0.544
VERIDICALITY $\times$ [VOICE: passive]	-0.51733	3.52427	-0.147	0.883
VERIDICALITY $\times$ SLIFT	-0.49853	0.73560	-0.678	0.498

Table 1: Experiment 2: Results of the linear logistic regression model. Multiple R^2 : 0.3784, Adjusted R^2 : 0.3655. F-statistic: 29.44 on 11 and 532 DF.

Exp 2: Discussion

- Contrary to our expectation, our analysis revealed no effect of **veridicality**, **slift** or interactions among these two factors for compatibility with *which*-questions.
- **Possible methodological problems:**
 1. The correlation between **slift** and **preferentiality** isn't perfect (White et al. 2018)
 2. MegaAttitude only contains verbs and **lacks adjectives** that constitute a subset of relevant predicates (e.g., *glad*).
 3. The acceptability ratings (White & Rawlins 2016) and the veridicality ratings (White & Rawlins 2018) were measured with different sets of participants. This is important in view of Chemla et al.'s (2011) finding that a subject's judgment about NPI-licensing is affected by **their own** judgment about DE-ness.

Summary and prospects

- A close look at a particular account of this form, focusing on the **hope-wh* puzzle (non-veridical preferentials ~ anti-roгатivity).
- Preliminary experimental evidence.
- **Potential counterexamples**
 - Negative attitudes
 - ...
- **Methodological challenges**
 - Difficulty in operationalizing some of the semantic properties (e.g., preferentiality, focus sensitivity)
 - Scale of the data—within English; and across languages
 - Between-subject vs. within-subject measures

Appendix

Exp 1: Analysis

- A mixed-effects ordinal logistic regression was run on the responses in the interrogative block (**interrogative**) with the response in the veridicality block (**veridicality**) and whether the predicate involves *be* (**be**) and their interaction as fixed effects, and the random effects as in the following formula (in R, using package `ordinal`)

$$(33) \quad \text{interrogative} \sim \text{veridicality} * \text{be} \\ + (1 + \text{veridicality} * \text{be} | \text{Participant}) + (1 + \text{veridicality} | \text{Pred})$$

- **Rationale for including be:** Since the items don't include a preposition, there might be interaction between **veridicality** and **be**. Given veridicality, a predicate without *be* (e.g., *like*) is more acceptable with interrogative complements than one with *be* (e.g., *be surprised*).

Exp 1: Results

	β	Std. Error	z-value	p	
VERIDICALITY	0.22236	0.07020	3.167	0.00154	**
BE	0.30756	0.55548	0.554	0.57979	
VERIDICALITY $\times$ BE	-0.01072	0.11401	-0.094	0.92506	

Table 2: Experiment 1: Results of the ordinal mixed-effects model. The standard deviation of the Subject random intercepts was .76 and that of the Predicate random intercepts was .87. Significance levels: .05 (*), .01 (**).